

Treasury Products — Quick Reference Card

Universal pricing recipe · 33 products · risk metrics · ALM patterns. Deep companion: treasury_products_cheatsheet.pdf

◆ Universal Symbols

N	notional (domestic)
T, τ	maturity / year fraction
K	strike / contracted rate (LOCKED)
DF(t)	exp(−z(t) · t)
DF(t,s)	exp(−(z(t)+s/1e4) · t)
S, fx	FX spot (dom/foreign)
b	breakeven inflation
A(T)	Σ τ · DF(t_i) (annuity)
σ, N(·)	vol, std normal CDF

Master skeleton: V = (market_implied − contract) · scale · discount

◆ Master Recipe (4 steps)

1. List cashflows — when, ccy, contingent on what

2. Substitute each random CF with risk-neutral expectation (↓)

3. Discount by DF(t) (or DF(t,s) for credit)

4. Sum: V = Σ E[CF_t] · DF(t)

SUBSTITUTION TABLE

Random CF	Replace with
floating rate L	(D ₁ /D ₂ − 1)/τ
spot FX at T	S · DF_f/DF_d
inflation index	(1+b) ^T
option max(S−K,0)	DF(T) · [F · N(d ₁) − K · N(d ₂)]
survival to t	Q(t) = exp(−λt)
default in (t ₁ ,t ₂]	Q(t ₁) − Q(t ₂)
swap rate (CMS)	fwd + convexity adj.

◆ Universal Patterns

A — LINEAR-IN-STRIKE

V = (market − contract) · scale · disc

Par strike = market-implied (set V=0).

Used by: FX fwd, FRA, ZCIS, IRS, CDS

B — BASIS × ANNUITY

V = b · N · A(T) [· FX]

Used by: Sec lending, tenor basis, XCCY basis

C — CASHFLOW STRIP PV

V = ± Σ cf_i · uplift_i · DF(t_i, s)

Used by: Bond, FB, Linker, Loan, Annuity Liab

Recognise the pattern → recall the formula. Each product is one of these three shapes.

◆ Walk-through: FRA via recipe

1. CF: N(L−K)τ at T₂ 2. L → (D₁/D₂−1)/τ = fwd 3. discount: · DF(T₂) 4. V = N(fwd−K)τ · DF(T₂)

Every entry below is a 1-line output of the same recipe.

◆ Reference Curve (toy examples)

t	z(t)	DF(t)
1y	4.50%	0.95600
2y	4.40%	0.91576
3y	4.30%	0.87897
5y	4.20%	0.81058
10y	4.10%	0.66365

A(5y) = Σ τ · DF(1..5) = 4.40498

◆ Portfolio Recipe (ALM)

1. Exposure vector — bucket by KRD (2y/5y/10y/30y), IE01, CS01, FX Δ, vega

2. Aggregate signed contributions (assets +, liabs −)

3. Hedging set — IRS strip + Linker strip + nominal bonds + OIS + ZCIS

4. Solve linear system → hedge notionals; report residuals

Parallel DV01 ≠ KRD. A 10y liab hedged with 5y IRS is parallel-neutral but long-end exposed. KRD-mismatch = canonical pension failure mode.

◆ Convention reminders

• bp → decimal: / 1e4. 15bp = 0.0015.

• Inflation compounds: (1+b)^t (NOT 1+b·t).

• τ must match contract day-count.

• Annuity liability is NEGATIVE.

• FRA fwd: SIMPLE (D₁/D₂−1)/τ, not continuous.

• Swap float telescope: 1 − DF(T) (single-curve).

• CDS hazard: λ ≈ s/(1−R); Q(t)=exp(−λt).

• Black option vol: lognormal in F, not S.

• Multi-curve (post-2008): OIS discount + LIBOR/SOFR project — drills ignore.

● I — Cash & Money Market

FX Spot (§1)

| V = N_for · S · fx_factor

Own foreign currency. Pure FX delta; no rate or time.

Risk: FX

Repo / Reverse Repo (§2)

| V = N(1 + r · T) · DF(T) − N

Secured short-term borrow (repo) / lend (reverse). V≈0 at par; small DV01.

Risk: rate (small)

Sec Lending (§3)

| V = (fee_bp/1e4) · N · A(T)

Lend securities; receive fee per period. Pattern B.

Risk: rate (tiny)

Amortising Loan (§4, linear amort)

| V = Σ cf_i · DF(i, spread_bp)

| amort = P/n, cf_i = amort + c · outstanding

Equal principal/period + interest on remainder. For annuity-style (equal total pmt) use pmt = P · c / (1 − (1+c)^{−n}).

Risk: rate, credit spread (CS01)

E1 T-Bill / discount paper

| V = face · DF(T) (zero coupon, govt)

| dy = (face−P)/face · 360/days

Risk: rate

E2 Commercial Paper

| V = face · DF(T, spread_bp)

Short corp/bank discount paper.

Risk: rate, credit

▲ II — Forwards

FX Forward (§5)

| V = N_for · (S · fx_factor − K) · DF(T)

Locked-rate currency exchange at T. K_par = S · DF_f/DF_d (IRP).

Risk: FX, rate

FRA (§6)

| fwd = (DF(T₁)/DF(T₂) − 1)/τ

| V = N · (fwd − K) · τ · DF(T₂)

Locked rate on future deposit [T₁,T₂]. K_par = fwd.

Risk: rate

MTM uses curve-implied fwd; real settle = N(L−K)τ/(1+L · τ) paid at T₁.

■ III — Bonds (Pattern C — cashflow strips)

Vanilla Bond (§7)

| V = Σ cf_i · DF(t_i, spread_bp)

Coupons + principal, discounted at issuer's credit curve.

Risk: rate (DV01), credit (CS01)

Foreign Bond (§8)

| V = (Σ cf_i · DF(t_i, s_bp)) · S · fx_factor

Bond in foreign currency, translated at spot.

Risk: rate, credit, FX

Linker (inflation-linked) (§9)

| V = Σ cf_real_i · (1+b)^{t_i} · DF(t_i)

CFs uplifted by inflation. Long inflation = b ↑ → V ↑.

Risk: rate, breakeven (IE01)

Indexation lag: 3M (Gilts) / 8M (TIPS) — adds small IE01 basis.

E3 Zero-Coupon Bond (long)

| V = face · DF(T, s)

| D_mac = T; D_mod = T / (1 + y/m)

No coupons; max duration per \$ of face.

E4 Floating-Rate Note (FRN) (at reset)

| V (at fixing) ≈ face · spread · A + face

Resets to ~par; duration ≈ time-to-next-fixing. Between resets the locked coupon adds accrual.

Risk: spread (dominant), tiny rate

E5 Callable Bond

| V = V_straight − V_call_option

Issuer's right to redeem early → bondholder bears asymmetric risk; negative convexity.

Risk: rate, credit, vol

E6 Convertible Bond

| V = V_straight + V_equity_call

Bond + long call on issuer's equity. Multi-factor.

Risk: rate, credit, equity, equity vol

◆ IV — Swaps

Interest Rate Swap (IRS) (§10)

| $V_{\text{rec}} = N \cdot [c \cdot A(T) - (1 - DF(T))]$
 where $A(T) = \sum \tau \cdot DF(t_i)$
| $c_{\text{par}} = (1 - DF(T)) / A(T)$

Exchange fixed-c vs floating; receive-fix sign shown. Pay-fix flips sign.

Risk: rate (THE rates instrument)

ZCIS Inflation Swap (§11)

| $V = N \cdot DF(T) \cdot [(1+b)^{\Delta T} - (1+K)^{\Delta T}]$
| $K_{\text{par}} = b$

Single payment at T: inflation index vs fixed.

Risk: rate, breakeven

Tenor Basis Swap (§12)

| $V = b \cdot N \cdot A(T)$
| $b = (\text{basis_bp} + \text{shock_bp})/1e4$

e.g. 3M vs 6M floating spread. Pattern B.

Risk: rate, tenor-basis

XCCY Basis Swap (§13)

| $V = N \text{ for } \cdot S \cdot \text{fx_factor} \cdot b \cdot A(T)$

Cross-currency basis leg in foreign ccy; FX sensitive.

Risk: rate, xccy-basis, FX

| Real XCCY: principal exchange + MtM resets (drill omits).

Total Return Swap (TRS) (§14)

| $V = \text{PV}(\text{reference asset}) - \text{funding_notional}$

Synthetic long the reference (balance-sheet light).

Risk: inherits reference's risks

E7 Overnight Index Swap (OIS)

| Same as IRS; float leg = compounded 0/N

Risk-free swap; benchmark discount curve post-2008.

E8 CMS

| float leg = swap rate (e.g. 10y) → convexity adj. needed (SABR surface dependent)

Bet on curve shape (steepening/flattening).

E9 Asset Swap (ASW)

| Bond + offsetting IRS → synthetic floater

| ASW spread ≈ bond coupon - par swap rate

Isolates credit spread, neutralises rate.

E10 YoY Inflation Swap

| Each year: $N \cdot (\pi_i - K) \cdot \tau$

| (vs ZCIS = single compounded payment)

Hedge year-by-year inflation; convexity adj vs ZCIS.

▲ V — Credit

Credit Default Swap (CDS) (§15)

| $\lambda = \text{market_spread} / (1-R)$
| $Q(t) = \exp(-\lambda \cdot t)$
| $\text{premium} = \sum \tau \cdot DF \cdot Q$
| $\text{protect} = \sum (Q_{\text{prev}} - Q) \cdot DF$
| $V_{\text{buyer}} = N \cdot [\text{protect} \cdot (1-R) - \text{premium} \cdot K]$

Buyer pays K, receives (1-R)·N on default. Buyer wins when spread widens.

K_par = market_spread.

Risk: credit spread (dominant), rate (small)

E13 CDS Index (CDX/iTraxx)

| Same as §15. Each defaulting name pays (1/n)·(1-R)·N.

| Remaining notional continues to accrue premium.

CDX IG = 125 US IG names @ 100bp; HY @ 500bp. NOT a first-to-default basket.

Risk: spread (CS01), correlation (tranches)

E14 MBS — agency pass-through

| $V = \sum E[\text{CF}_t \mid \text{prepay_model}] \cdot DF(t, \text{OAS})$
| (MC over rate paths + CPR/PSA prepayments)

Negative convexity from prepayment option.

Risk: rate (effective DV01), prepay, OAS, vol

● VI — Options

FX Option (§16)

| $F = S \cdot \text{fx_factor} / DF(T)$
| $d1 = [\ln(F/K) + \frac{1}{2}v^2T] / (v\sqrt{T}); \quad d2 = d1 - v\sqrt{T}$
| $V_{\text{call}} = N \cdot DF(T) \cdot [F \cdot N(d1) - K \cdot N(d2)]$
| $V_{\text{put}} = N \cdot DF(T) \cdot [K \cdot N(-d2) - F \cdot N(-d1)]$

Right to exchange currencies at K on T.

Risk: FX (delta), vol (vega), rate

Swaption (§17)

| $A = \sum \tau \cdot DF(t_i) \quad (\text{swap annuity})$
| $S = [DF(T_{\text{exp}}) - DF(T_{\text{end}})] / A \quad (\text{fwd swap rate})$
| d1, d2 as Black on S
| $V_{\text{payer}} = N \cdot A \cdot [S \cdot N(d1) - K \cdot N(d2)]$
| $V_{\text{recv}} = N \cdot A \cdot [K \cdot N(-d2) - S \cdot N(-d1)]$

Option to enter IRS at K. Black model on forward swap rate.

Risk: rate, vol (vega dominates at ATM)

E11 Cap / Floor / Caplet

| Caplet at t_i: $\tau \cdot N \cdot \max(L_i - K, 0)$ paid at t_{i+1}

| Cap = $\sum \text{caplets}$, each Black on fwd rate

Hedge floating-rate exposure. Pure vol play.

E12 Bermudan Swaption

| Multiple exercise dates → LSMC / lattice

| V > European by "early exercise premium"

Used for callable bond hedging.

Risk: rate, vol, mean reversion (model dep.)

E15 Variance Swap

| Payoff = $N \cdot \text{var} \cdot (\sigma^2_{\text{real}} - K^2)$

| N_vega = $2K \cdot N \cdot \text{var}^-$ (approx near K)

| Replication: OTM calls + puts (1/K² weights)

Pure realised-vol exposure; tail hedge.

E16 Equity TRS / E17 Commodity Fwd

| Equity TRS: $V = \text{PV}(\text{equity total return}) - \text{funding}$

| Commodity Fwd: $V = N \cdot (F_T - K) \cdot DF(T)$

Synthetic equity / hedge raw materials.

■ VII — Liabilities

Annuity Liability (§18)

| $V = - \text{payment} \cdot \sum (1+b)^{\Delta t} \cdot DF(t), \quad t=1..years$

Pension-style: you OWE inflation-linked payments. NEGATIVE PV.

Risk: rate (rates ↓ → V more neg), breakeven (b ↑ → V more neg).

Hedge: Linker matches IE01 partly; IRS matches DV01.

Sign trap: liability has opposite-sign DV01 to a long bond. Pick one DV01 convention across your stack. KRD-mismatched hedges (parallel-neutral, long-end-exposed) are the canonical pension hedge failure.

◆ VIII — Risk Metrics

Macaulay Duration (years)

| $D_{\text{mac}} = \sum t_i \cdot \text{PV}_i / \sum \text{PV}_i$

PV-weighted average time to cashflows.

Modified Duration (%)

| $D_{\text{mod}} = D_{\text{mac}} / (1 + y/m)$

| $\Delta P/P \approx -D_{\text{mod}} \cdot \Delta y$

DV01 / PV01 (\$)

| $DV01 = D_{\text{mod}} \cdot P \cdot 0.0001$

\$ P&L per 1 bp rate move.

Convexity

| $C = \sum t_i (t_i + 1/m) \cdot \text{PV}_i / [P \cdot (1 + y/m)^2]$

| $\Delta P/P \approx -D_{\text{mod}} \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$

Positive for long vanilla bonds → cushions losses, boosts gains.

Greeks (Black on forward F)

Δ delta	$DF(T) \cdot N(d1)$
Γ gamma	$DF(T) \cdot \phi(d1) / (F \cdot \sigma \cdot \sqrt{T})$
v vega	$DF(T) \cdot F \cdot \phi(d1) \cdot \sqrt{T}$
Θ theta	– for long opt (time decay)
ρ rho	call-rho > 0

| Gamma uses **F**, not S, in the denominator.

Numerical (FD) when no closed-form

| $DV01_{\text{fd}} = [V(+1bp) - V(-1bp)] / 2$

| $\text{Conv}_{\text{fd}} = [V(+h) - 2V + V(-h)] / (h^2 \cdot V)$

◆ Risk-factor sign cheatsheet

Long position in...	DV01	IE01	FX Δ
Bond / Loan / Linker	–	+ (Linker only)	0
IRS receive-fixed	–	0	0
IRS pay-fixed	+	0	0
ZCIS receive-infl	≈0 at par	+	0
FX Spot / Fwd (long foreign)	+	0	+
Annuity Liability (short)	+	–	0
FX/Swaption (long call)	small	0	+

Convention: signs of ∂V/∂r (V rises when factor rises = +). Trader DV01 = –∂V/∂r just flips signs uniformly. Receive-fix = long duration ⇒ same sign as long bond.

◆ Top ALM caveats (full list in gold-ref PDF)

- **Multi-curve**: OIS discount + LIBOR/SOFR projection (drills: single-curve).
- **LDI regimes** (UK FRS102 / US IAS 19): liab discounted on AA-corp → swap-vs-AA basis is unhedged.
- **LPI caps**: UK pension inflation capped at 5% → liability non-linear; ZCIS doesn't hedge convexity.
- **Sept 2022 gilt crisis**: LDI hedge collateral calls cascaded — the hedge itself has liquidity risk.
- **Fisher**: nominal ≈ real + breakeven. Linker DV01 = REAL-rate sensitivity, not nominal.